

AP CSP | Lesson 04: Test & Improve

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B is correct: $00101101_2 = 32 + 8 + 4 + 1 = 45$. A, 43, corresponds to 00101011. C, 46, corresponds to 00101110. D, 53, corresponds to 00110101. Each incorrect value assigns different positional weights; leading zeros do not change the value.

Q2. Answer: C

C is correct: the visitor's selection supplies data to the program, and the displayed value sends data to a device. A mistakes processing for input and reverses the visitor selection's role. B names an output device as input and mistakes internal program state for output. D reverses the displayed result's role and treats internal computation as output.

Q3. Answer: B

B is correct: y receives $4 + 2 = 6$ when the second statement runs. Changing x later does not automatically recalculate y . A ignores the addition. C substitutes x 's final value for y . D incorrectly treats y as a permanent formula tied to the current value of x .

Q4. Answer: D

D is correct: I breaks the language grammar, while II is a fault reported during execution under the stated language rules. A misclassifies both situations. B also misclassifies both. C identifies the syntax error correctly but overlooks the explicit execution-time fault in II. The example does not claim that every language handles division by zero identically.

Q5. Answer: A and C

A and C are correct. For 2 tickets, the faulty output is 30 but the required total is 27. For 50 tickets, the faulty output is 750 but the required total is 603. B does not expose the fault: both calculations give 15 for one ticket. D is outside the stated valid domain; 0 must not be labeled a valid test input. Award one mark only for A and C with no other selections.

Q6. Answer: A

A is correct: it uses specific user evidence to revise the interface and checks the effect of the revision. B ignores usability and confuses syntactic validity with meeting user needs. C changes a correct calculation and would charge the fee twice. D excludes the visitor's perspective and ends testing without checking the reported problem.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) Six bits: $2^6 = 64 \geq 41$, but $2^5 = 32 < 41$. (b) 64 patterns. (c) 63.

1 mark: Gives six bits and shows that $2^6 = 64$ is sufficient for 41 exhibits.

1 mark: Shows that one fewer bit fails: $2^5 = 32 < 41$.

1 mark: States that six bits provide 64 distinct patterns.

1 mark: States that the largest represented nonnegative integer is 63.

Q8. 4 marks

Answer: (a) A program is integral to the app's translation function. (b) Input: the visitor's recorded speech/audio data; output: the displayed translated text. (c) Increase or allow adjustment of the displayed text size, then ask the visitor to read a translated example and observe whether it is readable.

1 mark: Explains that a program is integral to the app's translation function.

1 mark: Identifies the visitor's speech/audio data as input.

1 mark: Identifies the displayed translated text as output.

1 mark: Suggests a relevant text-size/readability change and a user check tied to the reported difficulty.

Q9. 4 marks

Answer: (a) $a = 4$, $b = 7$. (b) $a = 10$, $b = 9$. (c) 10, then 9. (d) The earlier assignment computed $a + 3$ using a 's value at that time and stored 7 in b ; it did not create a continuing relationship.

1 mark: Gives both $a = 4$ and $b = 7$ after the second statement.

1 mark: Gives both $a = 10$ and $b = 9$ after the fourth statement.

1 mark: Gives output in the correct order: 10, then 9.

1 mark: Explains that assignment stores the expression's value at that time; b is not automatically recomputed when a changes.

Q10. 4 marks

Answer: (a) $\text{perimeter} \leftarrow 2 * (\text{length} + \text{width})$, or an equivalent expression. (b) 22. (c) The original computes only $\text{length} + \text{width}$, giving 11 instead of the full perimeter. It runs but does not meet the requirement. (d) It takes a rectangle's length and width and displays the total distance around its four sides.

1 mark: Provides a correct assignment such as $\text{perimeter} \leftarrow 2 * (\text{length} + \text{width})$ or $\text{perimeter} \leftarrow \text{length} * 2 + \text{width} * 2$.

1 mark: Gives corrected output 22.

1 mark: Explains that the original calculation omits the second pair of sides, producing 11 instead of 22, despite valid syntax.

1 mark: States that the program uses length and width as inputs to display the rectangle's full perimeter.

Q11. 4 marks

Answer: (a) $1 \rightarrow 15$ and $50 \rightarrow 603$. (b) Expected 27; faulty actual 30. (c) The parentheses make the program multiply the booking fee by the number of tickets, so it charges the fee per ticket. (d) $\text{total} \leftarrow \text{tickets} * 12 + 3$.

1 mark: Gives both correct boundary totals: 15 for 1 and 603 for 50.

1 mark: Gives both expected 27 and faulty actual 30 for input 2.

1 mark: Explains that the faulty expression multiplies the fee by tickets instead of adding it once per order.

1 mark: Writes $\text{total} \leftarrow \text{tickets} * 12 + 3$ or an equivalent correct assignment.

Q12. 4 marks

Answer: (a) The navigator can predict the result for a test input, trace the calculation, or compare the code with the per-order fee requirement and explain findings to the driver. (b) Add a clear label such as "Total including booking fee:". (c) Ask the visitor to interpret the revised display and rerun the numeric tests to confirm the totals remain correct. (d) "The input lines were adapted from the teacher's Chapter 1 Programming Lab starter. [Student name] adapted those lines; [names] developed the calculation and output changes." Credit only the contributions actually made.

1 mark: Gives a specific active navigator contribution, such as independently predicting a test result or tracing and discussing a code step.

1 mark: Proposes a clear output label or equivalent change that explicitly addresses whether the fee is included.

1 mark: Includes both a check of the visitor's understanding and rerunning numeric tests after the revision.

1 mark: Acknowledges that the input lines were adapted from the teacher's starter/Chapter 1 Programming Lab, identifying source and affected code.